Module	Problem (Variables: n, m, V, E, W)	Candidates (with Complexity)	Selection (Primary vs Baseline)	Justification (Data Property, Assumptions & Failure Case)
A. Relief inventory	**0/1 Knapsack**. Input: n crates with weights w_i in [1, 20] kg, values v_i in [10, 100], truck capacity W <= sum(w_i). Output: Set S subset {1..n} maximizing total value sum(v_i) subject to total weight sum(w_i) <= W.	1. **1D Tabular DP**: O(n * W) pseudo-poly. 2. **Branch and Bound**: O(2^n) worst-case. 3. **Greedy by v/w ratio**: O(n log n).	**Primary:** 1D Dynamic Programming O(n * W). **Baseline:** Greedy value-density heuristic O(n log n).	Weights and capacity are strictly positive integers with W <= 30,000, where pseudo-polynomial DP runs in milliseconds. Assumes static integer crate weights. Fails if weights/capacities are continuous non-integral reals or capacity W >> 10^7.
B. Route planning	**Single-Source Shortest Paths & Reachability**. Input: Directed graph G=(V, E) with road traversal time minutes(e) in [1, 60] and destination penalty(v) in {0,1,2,4,6}. Output: Cheapest path minimizing sum(minutes + penalty_v), reachable set, and min-hop path.	1. **Dijkstra with Min-Heap**: O((V + E) log V). 2. **Bellman-Ford**: O(V * E). 3. **Unweighted BFS**: O(V + E) for hop count.	**Primary:** Dijkstra with Min-Heap O((V+E)log V) and BFS O(V+E). **Baseline:** Bellman-Ford O(V * E).	Minutes and penalties are non-negative (edge cost >= 1), perfectly satisfying Dijkstra's greedy non-negative edge invariant. Assumes static road state per day. Fails if negative cost edges or negative penalty-relief cycles exist.
C. Rescue scheduling	**Interval Scheduling & Weighted Interval Scheduling**. Input: m rescue windows [start_i, finish_i] with values v_i. Output: Maximum cardinality compatible set (touching allowed) or maximum value compatible set.	1. **Greedy Earliest Finish Time (EFT)**: O(m log m). 2. **Weighted Interval DP with Bisect**: O(m log m). 3. **Integer Linear Program**: O(2^m).	**Primary:** Greedy EFT for count; DP + Binary Search for value O(m log m). **Baseline:** Recursive search O(2^m).	Intervals are 1D continuous segments [0, 1440]. Sorted finish times guarantee the greedy-choice property for count and optimal substructure for weighted DP via binary search predecessor lookup. Fails if windows require multi-crew matching or have setup dependencies.
D. Emergency queue	**Dynamic Priority Queue Simulation**. Input: A alarms arriving at non-decreasing tick t_i with priority p_i in [1, 10] (10 most urgent). Output: Release order answering one alarm per tick in urgency order among available alarms.	1. **Dial's Bucket Queue / Multi-level Array**: O(A + K) where K=10. 2. **Binary Max-Heap**: O(A log A). 3. **Unsorted Scan**: O(A^2).	**Primary:** Dial's Multi-level Bucket Queue O(A) total, O(1) amortized. **Baseline:** Binary Max-Heap O(A log A).	Priority values are small, discrete integers bounded in [1, 10]. Direct array buckets permit O(1) insertion and O(1) extraction of the highest-priority incident. Fails if priorities are arbitrary continuous real numbers or priority space K >> A.
E. Radio logs	**Exact Pattern Search, String Edit Distance & Data Compression**. Input: Alphabet Sigma (|Sigma|=38), log length N, reports of lengths L_a, L_b. Output: All phrase occurrences, min edit script (ins/del/sub), and optimal prefix-free code.	1. **Knuth-Morris-Pratt (KMP)**: O(N + M) search. 2. **Wagner-Fischer DP**: O(L_a * L_b) edit dist. 3. **Huffman Greedy Tree**: O(|Sigma| log |Sigma|).	**Primary:** KMP search O(N+M), Wagner-Fischer DP O(L_a L_b), Huffman O(|Sigma| log |Sigma|). **Baseline:** Naive search O(N*M).	Fixed discrete alphabet (|Sigma|=38) and static log text allow linear deterministic automata / failure function preprocessing. Character frequencies in logs exhibit strong skewness (e.g. Study 6 and Trap 49), guaranteeing Huffman optimality. Fails under wildcard / regex stream queries.
F. Network planning	**Minimum Spanning Tree (MST)**. Input: Connected reachable graph G=(V, E) with cable costs c_e in [1, 100]. Output: Minimum cable cost acyclic subgraph connecting all reachable locations.	1. **Kruskal with DSU & Counting Sort**: O(E * alpha(V)). 2. **Prim with Binary Heap**: O(E log V). 3. **Boruvka's Algorithm**: O(E log V).	**Primary:** Kruskal with Disjoint Set Union (DSU) O(E * alpha(V)). **Baseline:** Prim's Algorithm O(E log V).	Road network is relatively sparse (E <= 3V) and cable costs are integers in [1, 100]. Edge weights can be sorted in linear O(E) time using Counting Sort, reducing Kruskal to nearly O(E). Fails on dense graphs where E = Theta(V^2) and Prim with dense matrix is O(V^2).
G. Chains and cuts	**Matrix Chain Multiplication (MCM) & Rod Cutting**. Input: n pipeline stage dimensions; drum length L with price array P[1..L]. Output: Optimal associative parenthesization minimizing scalar multiplications; optimal cutting maximizing total revenue.	1. **Bottom-up Dynamic Programming**: O(n^3) MCM, O(L^2) Rod Cut. 2. **Memoized Recursion**: O(n^3) MCM, O(L^2) Rod Cut. 3. **Greedy Heuristic**: O(n log n) MCM.	**Primary:** Tabular Dynamic Programming O(n^3) MCM, O(L^2) Rod Cut. **Baseline:** Greedy dimensional / density heuristic.	Both problems possess strict optimal substructure and overlapping subproblems without cyclic dependencies. Stage counts (n <= 300) and drum lengths (L <= 2000) evaluate rapidly under tabular DP. Fails if drum cutting incurs cutting kerf material loss or stage operators are non-associative.
H. Hard planning questions	**NP-Hard Combinatorial Decision Problems**: 3-SAT (Safety), Exact Subset Sum (Rations), Metric TSP (Tour), and Minimum Vertex Cover (Sensors). Output: Exact satisfying assignment, exact target subset, shortest tour, and min cover.	1. **CDCL / DPLL with Unit Propagation** for 3-SAT. 2. **Meet-in-the-Middle / DP**: O(2^(n/2)) or O(n * Target). 3. **Held-Karp DP** O(n^2 2^n) / **2-Opt**. 4. **FPT Branch & Bound**: O(1.2738^k + kn).	**Primary:** CDCL for 3-SAT, Meet-in-the-Middle for Rations, Held-Karp (Tier S) / 2-Opt (L/X) for TSP, FPT B&B; for MVC. **Baseline:** Brute-force O(2^n) / O(n!).	Planted solutions exist by construction (planted cover k <= n/2, planted assignment, planted subset). Small parameter sizes on Tier S (n <= 15) allow exact Held-Karp / FPT algorithms to verify claims. Fails on unconstrained instances beyond the 3-SAT phase transition ratio (rules/vars ~ 4.267).
I. Nearest sites and checksums	**2D Closest Pair of Points & Arbitrary-Precision Multiplication**. Input: n coordinates (x, y) in [0, 999]^2; two long decimal numbers from concatenated crate serials (d digits). Output: Minimum squared Euclidean distance; high-precision product.	1. **Divide and Conquer Closest Pair**: O(n log n). 2. **Karatsuba Multiplication**: O(d^1.585). 3. **Naive Pairwise & Schoolbook**: O(n^2) and O(d^2).	**Primary:** Divide-and-Conquer Closest Pair O(n log n) and Karatsuba O(d^1.585). **Baseline:** Pairwise distance O(n^2) and Schoolbook multiplication O(d^2).	Sites lie in a bounded Euclidean plane where recursive geometric strip pruning guarantees at most 7 candidates per point; serial digit lengths d reach thousands of digits in Tier L where Karatsuba achieves substantial sub-quadratic speedup. Fails for points in high dimensions (D >> 10).

Member	Contribution (Iteration 1 Deliverables & Proposed Iteration 2 Modules)
Ch Faizan Zahid **Roll No: 22I-0533** **Section: B**	Iteration 1: Generator architecture, knapsack/claims exact solver (crates, claims sheets), graph generator & closures (roads, waves sheets), validator core engine (V1-V5 checks), approach note part 1 & 2. Proposed Iteration 2 Modules: Module A (Relief Inventory), Module B (Route Planning), Module H (Hard Planning Questions: 3-SAT & TSP).
Mohid Abbas **Roll No: 22I-0074** **Section: B**	Iteration 1: Incident & rescue window generators (incidents, windows sheets), radio log & report generators (radio_log, radio_reports sheets), validator checks (V6-V7 signature & variety checks), validator test suite (broken test cases), approach note part 3 & 4. Proposed Iteration 2 Modules: Module C (Rescue Scheduling), Module D (Emergency Queue), Module E (Radio Logs).
Fizan Tariq **Roll No: 22I-0547** **Section: B**	Iteration 1: Geometric sites & sensor network generators (sites, tour, sensor_edges, sensor_cover sheets), pipeline & drum generators (pipeline_dims, drum_prices, rations sheets), validator trap checks (V8 checks), validation report compilation, repository & packaging. Proposed Iteration 2 Modules: Module F (Network Planning), Module G (Chains and Cuts), Module I (Nearest Sites and Checksums).